

Total No. of pages: 2

Roll No.:

**B.TECH.- FOURTH SEMESTER
MID-SEMESTER EXAMINATION, February 2026**

Course Code-CMMTC13

Course Title-Applied Linear Algebra

Time: 1:30 hours

Max Marks-25

Note: Attempt all questions in given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
1(a)	Check whether the set of integers under subtraction is a group or not.	2.5	CO1
1(b)	Show that the set of all $n \times n$ real matrices whose diagonal is zero, is a vector space over $\mathbb{R}$.	2.5	CO1
2(a)	Let V be the space of functions $f : \mathbb{R} \rightarrow \mathbb{R}$. Check if $W = \{f(x) : f(0) = f(1) = 0\}$ is a subspace of V or not.	2.5	CO1
2(b)	Show that $W_1 \oplus W_2 = \mathbb{R}^2$, where $W_1 = \{(x, 2x) : x \in \mathbb{R}\}$ and $W_2 = \{(x, 3x) : x \in \mathbb{R}\}$.	2.5	CO1
3(a)	Is the set A linearly independent in $C(\mathbb{R})$? $A = \{\sin x, \sin 2x, \sin 3x, \sin^3 x\}$.	2.5	CO1
3(b)	Show that the linear transformation $P : V \rightarrow V$ such that $P(x, y) = (x, 0)$ is a projection and find its range and kernel.	2.5	CO2
4(a)	Verify rank-nullity theorem for the linear transformation $T : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ such that $T(x_1, x_2, x_3, x_4) = (x_1 + x_4, x_2 + x_3, x_1 + x_2)$.	2.5	CO2
4(b)	Let $T : M_{2 \times 2}(\mathbb{R}) \rightarrow M_{2 \times 2}(\mathbb{R})$ be the linear transformation defined by $T \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a+b & b \\ c & c+d \end{pmatrix}$. Find a basis for the image of T .	2.5	CO2